

over 64-bit lengths to support very large transfers, the ordering for a "next block counter" SHALL use modulo 2^32 integer arithmetic. Therefore, if the last Block Counter was 0xFFFF_FFFF, the next expected Block Counter would be 0x0000_0000.

NOTE

Since the data length of blocks does not need to exactly match the Max Block Size in **Block/BlockEOF** messages, it is application-dependent whether Block Counters can be used to statelessly track the offset within a span of initiated transfer. That is, the relationship: **current_offset = start_offset + (max_block_size * block_counter)** is true **only** if it is contextually known that this usage applies for a given transfer application. This relationship, is SHALL NOT be assumed, since it may not apply, such as when variable-sized blocks are being sent to optimize a given data transfer flow.

11.22.6.2. BlockQuery Message

The **BlockQuery** message is sent by the driving Receiver to the follower to request the next block of data. This message implies a **BlockAck** of the previous block if no **BlockAck** was explicitly sent. The block counter SHOULD start at 0 at the start of the transfer, and increment by one for each subsequent block.

In asynchronous transfers, no **BlockQuery** messages are sent.

The fields of the BlockQuery message are:

Table 117. BlockQuery message fields

Field	Size	Notes
Message ID: BlockQuery(0x10)		
BlockCounter	4 octets	Unsigned little-endian

11.22.6.3. BlockQueryWithSkip Message

The **BlockQueryWithSkip** message MAY be sent by a driving Receiver to the follower to request the next block of data, after skipping an amount forward within the stream. This message implies a **BlockAck** of the previous block if no **BlockAck** was explicitly sent.

This message is semantically equivalent to a **BlockQuery**, but with the following additions:

- Before the next Block is sent by the Sender, the cursor within the underlying data transferred by the Sender SHALL be advanced by **BytesToSkip** bytes.
- If, after skipping **BytesToSkip** bytes, the cursor reaches the end of the file, or beyond, then the next message from the Sender SHALL be a **BlockEOF** with empty contents. In other words, there SHALL be no error indicated when receiving a request to skip past the end of the transferable data.
- The amount of **BytesToSkip** MAY be a size that does not match the current transfer’s maximum block size.

This message enables seeking forward in BDX transfers without requiring the termination of a